Faculty of Computers
& Artificial Intelligence

Benha University

SuperCareerAI

AI-Powered Platform for Freelance Matching, Job Recommendations & Smart CV Optimization

A senior project submitted in partial fulfillment of the requirements for the degree of Bachelor of Computers and Artificial Intelligence.

"Information Systems" Program,

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DECLARATION

We hereby certify that this material, which we now submit for assessment on the program of study leading to the award of Bachelor of Computers and Artificial Intelligence in *(Bachelor degree)* is entirely our own work, that we have exercised reasonable care to ensure that the work is original, and does not to the best of our knowledge breach any law of copyright, and has not been taken from the work of others save and to the extent that such work has been cited and acknowledged within the text of our work.

Signed: ______________________________

Date:

ABSTRACT

The rapid growth of digital employment platforms and freelancing marketplaces has increased the complexity of career opportunity discovery and application preparation for individuals. Job seekers and freelancers are often required to search across multiple platforms, evaluate opportunities manually, and repeatedly customize their CVs or proposals, resulting in significant time consumption and inefficiency. In addition, many candidates face challenges related to Applicant Tracking Systems (ATS), where technically unsuitable CV formats can lead to rejection despite adequate qualifications.

This graduation project, titled **SuperCareer AI**, presents an AI-powered information system designed to support job seekers and freelancers in discovering suitable opportunities and preparing optimized application materials. The proposed platform integrates job recommendation and freelance matching functionalities within a unified system, leveraging artificial intelligence and natural language processing techniques to analyze user profiles, opportunity requirements, and market trends.

The system provides intelligent recommendations for job positions and freelance projects

The platform adopts a user-centered and transparent design, ensuring that users retain full control over application decisions while benefiting from intelligent assistance. By reducing time and effort, improving matching accuracy, and enhancing application effectiveness, SuperCareer AI aims to increase accessibility to employment opportunities and contribute to a more efficient and inclusive digital labor ma.

Sample Graduation Project Index

LIST OF FIGURES:

Chapter One

1 INTRODUCTION

1.1 BACKGROUND:

The global labor market has experienced profound changes in recent years due to digital transformation and the rapid growth of online recruitment and freelancing platforms. Organizations increasingly rely on digital tools to identify and attract qualified candidates, while individuals depend on online job boards and freelancing marketplaces to access employment and project-based opportunities. Platforms such as LinkedIn, Indeed, and Glassdoor have become central to traditional job searching, while freelancing platforms like Upwork, Freelancer, and Fiverr have expanded access to remote and project-based work.

Despite the availability of numerous platforms, both job seekers and freelancers face significant difficulties in efficiently discovering suitable opportunities and preparing effective application materials. The fragmentation of opportunities across multiple platforms, combined with increasing competition, has made the process time-consuming and cognitively demanding. As a result, many qualified candidates fail to reach appropriate opportunities, while employers receive large volumes of unsuitable or poorly prepared applications.

Recent advancements in artificial intelligence, particularly in Natural Language Processing (NLP) and Large Language Models (LLMs), have created new possibilities for improving recruitment-related processes. AI technologies can support intelligent matching between candidates and opportunities, assist in preparing tailored application materials, and enhance transparency in the recruitment workflow. However, existing solutions often address these challenges in isolation rather than through a unified and user-centered approach.

1.2 PROBLEM DEFINITION:

The current processes of job searching and freelance project acquisition suffer from inefficiency, fragmentation, and limited transparency. Job seekers and freelancers are required to manually search across multiple platforms, analyze opportunities individually, and repeatedly customize CVs or proposals for each application. This results in excessive time consumption and reduced productivity.

For traditional job seekers, the use of Applicant Tracking Systems (ATS) introduces additional complexity. Many candidates are rejected due to technical or formatting issues rather than lack of qualifications, yet they receive little or no feedback explaining these rejections. Consequently, candidates are unable to identify weaknesses in their applications or improve their strategies.

Freelancers face similar challenges, particularly the need to write customized proposals for each project under significant time pressure. Highly competitive marketplaces reward early submissions, forcing freelancers to choose between speed and quality. Moreover, opportunities are distributed across multiple platforms, increasing the likelihood of missed projects.

These challenges lead to broader market inefficiencies, including poor matching between talent and opportunities, increased recruitment costs, and barriers to entry for recent graduates, career changers, and individuals with limited resources. There is a clear need for a system that reduces these inefficiencies while maintaining user control and transparency.

1.3 OBJECTIVES OF THE PROJECT:

The primary objective of the **SuperCareer AI** project is to design and implement an intelligent assistant platform that supports both traditional job seekers and freelancers in discovering opportunities and preparing high-quality application materials efficiently.

The specific objectives of the project are as follows:

- To aggregate job and freelance opportunities from multiple online platforms into a unified system.
- To apply NLP-based techniques to intelligently match user profiles with relevant job positions and freelance projects.

- To assist users in generating tailored CVs and freelance proposals aligned with specific opportunity requirements.
- To improve ATS compatibility of generated CVs and reduce rejections caused by technical issues.
- To provide transparent and explainable matching results that help users understand recommendation logic.
- To reduce the time and effort required for opportunity discovery and application preparation while preserving full user control over submission decisions.

1.4 SCOPE OF THE PROJECT:

The scope of the **SuperCareer AI** project focuses on supporting job seekers and freelancers during the preparation stage of the application process. The system provides tools for opportunity discovery, intelligent matching, and application material generation.

The project includes:

- User registration, authentication, and profile management.
- Aggregation of job and freelance opportunities from external platforms in a read-only manner.
- Intelligent matching of opportunities based on user skills, experience, and preferences.
- AI-assisted generation and customization of CVs and freelance proposals.
- ATS compatibility analysis and optimization for CVs.
- Tracking of user preparation activities, such as viewed opportunities and generated materials.

The project explicitly excludes direct application submission, employer communication, payment processing, and portfolio hosting. All applications are completed by users on external platforms to ensure ethical compliance and user agency.

1.5 IMPORTANCE OF THE PROJECT:

The **SuperCareer AI** is important due to its potential impact on individuals, the labor market, and technological development.

For individual users, the platform enhances productivity by reducing the time spent searching for opportunities and preparing applications. By providing tailored materials and intelligent recommendations, users can focus on opportunities that better match their qualifications, improving application quality and success rates. The system also promotes learning by increasing user awareness of effective application practices and market demand for specific skills.

From a labor market perspective, improved matching between candidates and opportunities can reduce recruitment inefficiencies and lower the volume of unsuitable applications. This contributes to faster hiring processes and better utilization of human capital. Additionally, the platform supports accessibility by assisting individuals who may lack professional networks or access to career guidance resources.

Technologically, the project demonstrates the practical application of AI techniques in a real-world domain. It highlights how AI can augment human decision-making through transparent and user-centered design rather than replacing human judgment. Ethical considerations such as privacy, fairness, and explainability are integral to the system's design.

1.6 METHODOLOGY OVERVIEW:

The project follows an iterative development approach that emphasizes flexibility, continuous improvement, and user-centered design. The system is developed through successive phases, including requirements analysis, system design, implementation, testing, and evaluation.

Artificial intelligence techniques, particularly NLP and LLM-based content generation, are employed to support matching and document preparation tasks. Web-based technologies are used to implement a scalable and responsive platform, while security and privacy principles guide data handling and system architecture. Detailed descriptions of methodologies, algorithms, and technologies are presented in subsequent chapters.

1.7 DOCUMENT ORGANIZATION:

This graduation project report is organized as follows:

- **Chapter 1: Introduction** – Introduces the project's core concepts, defines the problem statement, and outlines the primary objectives, scope, and methodology.
- **Chapter 2: Literature Review and Market Analysis** – Examines the evolution of digital recruitment and AI in career development, identifies market gaps, and establishes the strategic positioning of the system.
- **Chapter 3: System Analysis** – Details the functional and non-functional requirements, identifies stakeholders, and provides a deep dive into the system's logic through various UML diagrams (Use Case, Activity, and Sequence diagrams) and data modeling (ERD , and the detailed database schema and Data Flow).
- **Chapter 4: System Design** –Focuses on the architectural framework, high-level design components.
- **Chapter 5: Implementation and Testing** – Covers the technical realization of the project, including coding environments and the evaluation of system performance against user requirements.
- **Chapter 6: Conclusion and Future Work** – Summarizes the project's achievements and suggests potential enhancements for future iterations.

This structure ensures logical progression from problem identification to solution design, implementation, and evaluation, in accordance with graduation project guidelines.

Chapter Two

LITERATURE REVIEW AND MARKET ANALYSIS

2.1 RELATED WORK:

Chapter Two is intended as a market analysis chapter, not a theoretical research chapter, which is why industry reports were essential to understand real-world hiring behavior

The recruitment and freelancing landscape has been significantly transformed by recent digital advancements. This section explores the evolution of recruitment and freelancing platforms and examines the role of artificial intelligence in shaping modern career development.

2.1.1 Evolution of Digital Recruitment Platforms Digital recruitment has become a cornerstone of modern labor markets. Organizations increasingly rely on Applicant Tracking Systems (ATS) to manage high volumes of applications. Industry reports indicate that over 90% of large companies utilize ATS to screen resumes before they reach human recruiters(cv-infinity.com). While these systems improve hiring efficiency for organizations, they create a significant barrier for candidates(fullsight.io). Many qualified applicants are filtered out due to formatting issues or lack of keyword optimization, highlighting a critical need for tools that ensure ATS compatibility(selectsoftwarereviews.com).

2.1.2 Freelancing Ecosystems and Competitive Pressure The gig economy has seen exponential growth, with research estimating over 163 million freelancer profiles globally across major marketplaces(arxiv.org). This massive influx of talent has intensified competition, leading to "Proposal Fatigue." Freelancers often spend a disproportionate amount of time drafting customized proposals for each project to stay competitive. Current platforms offer basic templates but lack intelligent, context-aware assistance that helps freelancers balance quality with the speed required in high-traffic marketplaces.

2.1.3 AI and NLP in Career Development The integration of Natural Language Processing (NLP) and Large Language Models (LLMs) has introduced new ways to enhance application materials. Studies have shown that AI-assisted resume generation can increase employment chances by approximately 8% by improving content quality and relevance(arxiv.org). However, generic AI tools often produce impersonal results. Effective career assistance requires domain-specific grounding to ensure that generated content is not only professional but also optimized for the specific technical requirements of recruitment algorithms.

2.2 COMPARISON AND ANALYSIS:

To identify the unique value of **SuperCareer AI**, it is essential to compare it with existing market solutions and identify the "Candidate-Side" gap.

Table 2.1: Comparison of SuperCareer AI with Existing Solutions

Feature	**Standard Job Boards (LinkedIn)**	**Freelance Platforms (Upwork)**	**Generic AI (ChatGPT)**	**SuperCareer AI**
Cross-Platform Aggregation	No	No	No	**Yes**
Dual Mode (Job + Freelance)	No	No	No	**Yes**
Structured ATS Feedback	Limited	No	No	**Yes**
Contextual Proposal Drafting	No	Basic Templates	Manual	**Automated & Custom**
Unified Application Tracking	Internal Only	Internal Only	No	**Centralized**

2.2.1 Identification of the Market Gap While digital recruitment platforms are technologically advanced, they often function as "black boxes" for the applicant. Candidates submit materials and receive little to no feedback on why they were rejected. Furthermore, the market is split between traditional employment and freelancing, which does not reflect the hybrid nature of modern professional careers where individuals often seek both types of opportunities.

2.2.2 Strategic Positioning of SuperCareer AI It is important to emphasize that **SuperCareer AI** does not compete with existing platforms like Upwork or LinkedIn. Instead, it acts as an **Intelligent Intermediary Assistant**. It complements these platforms by enabling users to discover opportunities more effectively and optimize their materials *before* submission. This approach fills a clear market gap without disrupting the existing recruitment ecosystem.

2.2.3 Evidence-Based Demand Market data supports the need for a solution like SuperCareer AI:

- **Application Barriers:** Over 60% of job seekers abandon applications due to technical complexity or repetitive manual entry(saasworthy.com)..
- **Feedback Vacuum:** 63% of applicants express a strong desire for feedback during the application process, even if they are not selected(saasworthy.com).
- **Quality Concerns:** Approximately 25% of purely AI-generated applications appear generic, reducing their success rate. SuperCareer AI addresses this by ensuring context-aware and personalized content(aljazeera.net).

2.3 SUMMARY OF FINDINGS:

The literature and market analysis confirm that the recruitment industry is fragmented and lacks user-centered optimization tools. The primary findings are:

1. **Inefficiency in Customization:** There is a lack of tools that bridge the gap between finding an opportunity and generating high-quality materials in one unified workflow.
2. **The ATS Barrier:** Pre-submission optimization is essential to overcome the silent filtering of resumes by automated systems.
3. **Fragmented Career Models:** Modern professionals need a tool that supports both traditional jobs and freelance projects simultaneously.
4. **Complementary Nature:** The project is positioned as a supportive assistant that enhances the effectiveness of existing platforms rather than replacing them.

By addressing these challenges, **SuperCareer AI** evolves beyond a simple search tool into a comprehensive career assistant that strengthens users' competitiveness in a saturated labor market.

Chapter Three

3 SYSTEM ANALYSIS:

3.1 SYSTEM OVERVIEW:

SuperCareer AI is an AI-powered information system designed to assist individuals in discovering suitable job opportunities and freelance projects while optimizing their application materials. The system acts as an intelligent assistant that aggregates opportunities from multiple external platforms, analyzes user profiles, and provides personalized recommendations along with AI-generated CVs and proposals.

The system does not replace existing recruitment or freelancing platforms; instead, it complements them by enhancing preparation and decision-making before application submission. SuperCareer AI focuses on the candidate side of the recruitment process and supports both traditional employment and freelancing within a unified workflow.

3.2 STAKEHOLDERS & USERS:

The stakeholders are individuals or entities that affect or are affected by the system.

Stakeholders:

- **Job Seekers & Freelancers**: The primary beneficiaries of the system's matching and generation capabilities.
- **System Administrators**: Responsible for maintaining the platform, managing data scraping, and monitoring AI performance.
- **Academic Supervisors**: Overseeing the technical and ethical implementation of the project.
- **External Platforms**: Indirect stakeholders (e.g., LinkedIn, Upwork) from which the system aggregates public data.

Users:

- **Job Seeker**
  - Searches for job opportunities
  - Generates and optimizes CVs
  - Receives job recommendations
- **Freelancer**
  - Searches for freelance projects
  - Generates project proposals
  - Receives project recommendations
- **System Administrator**
  - Monitors system performance
  - Manages users
  - Oversees data scraping and system operations

3.3 FUNCTIONAL REQUIREMENTS:

- The system shall allow users to create and manage a personal profile.
- The system shall aggregate job and freelance opportunities from multiple external platforms.
- The system shall analyze user profiles and match them with relevant opportunities.
- The system shall generate AI-powered customized CVs for job applications.
- The system shall generate AI-powered proposals for freelance projects.

- The system shall provide ATS compatibility analysis and optimization suggestions.
- The system shall notify users about new high-matching opportunities.
- The system shall allow administrators to monitor scraping activities and system health.

3.4 NON-FUNCTIONAL REQUIREMENTS:

Performance:

- The system shall be optimized for high responsiveness, ensuring that the user interface responds to requests within **2 seconds** under normal load conditions.
- Response times for AI-powered functions, such as CV generation or matching score calculation, shall be optimized to provide a seamless user experience.

Security:

- The system shall implement robust encryption protocols to protect sensitive user data, including personal information and uploaded resumes.
- The system shall employ secure authentication and authorization mechanisms (e.g., JWT or OAuth) to prevent unauthorized access to user accounts.
- All data transmissions between the client and the server shall be secured using industry-standard protocols (HTTPS/TLS).

Scalability:

- The system architecture shall be designed to support an increasing number of concurrent users and a growing database of job postings without any significant degradation in performance.
- The background scraping and AI processing modules shall be scalable to handle high volumes of data during peak recruitment periods.

Usability:

- The system shall provide an intuitive and user-friendly interface with a clear layout, ensuring that users can navigate through job discovery and document generation with minimal learning curve.
- The platform shall follow responsive design principles, ensuring full accessibility and consistency across various devices, including desktops, tablets, and smartphones.

Reliability and Availability:

- The system shall ensure high availability (aiming for 99.9% uptime), remaining accessible to users across different time zones.
- The platform shall be built with fault-tolerant mechanisms to ensure graceful recovery from unexpected system failures or downtime.

Maintainability:

- The system shall be developed using a modular architecture, allowing for efficient updates, bug fixes, and the seamless integration of new AI models or external job sources in the future.

Data Integrity:

- The system shall ensure data consistency and accuracy across the database, particularly regarding the synchronization between user profiles and generated application materials.

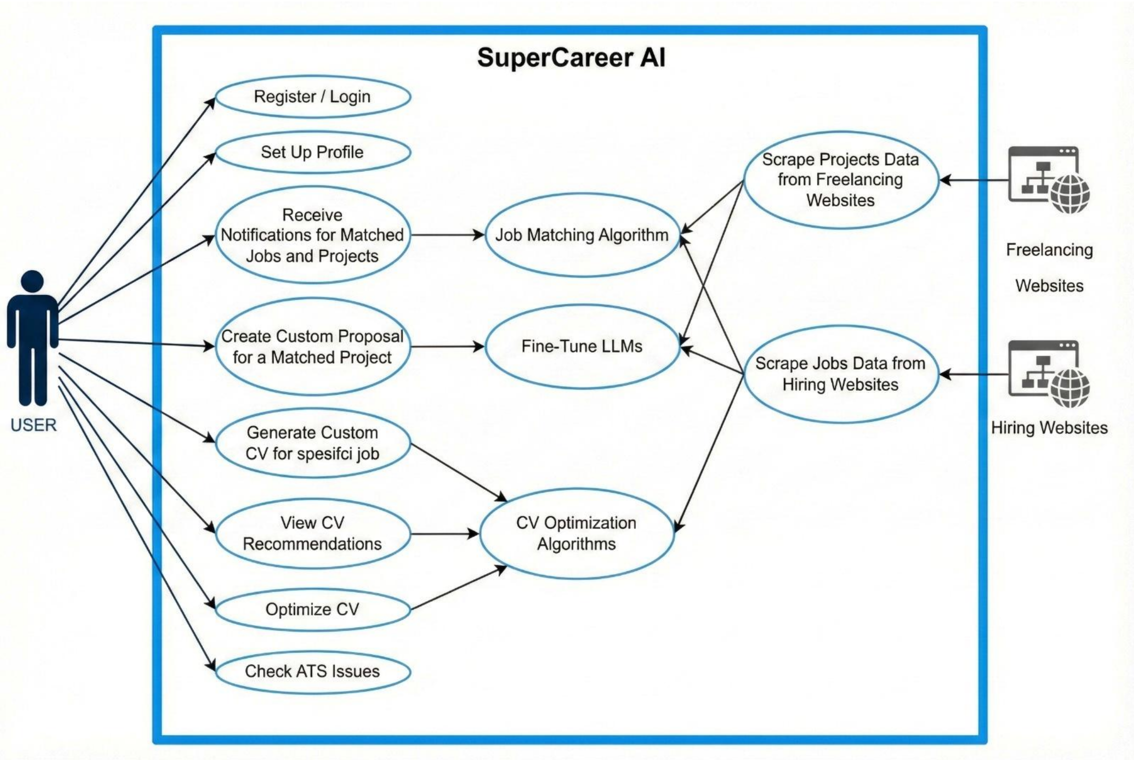
3.5 TOOLS & TECHNOLOGIES:

The selection of tools and technologies was based on scalability, maintainability, and suitability for AI-driven information systems.

- Frontend Technologies
- Backend Technologies
- Database

- AI & NLP Technologies
- Web Scraping Tools

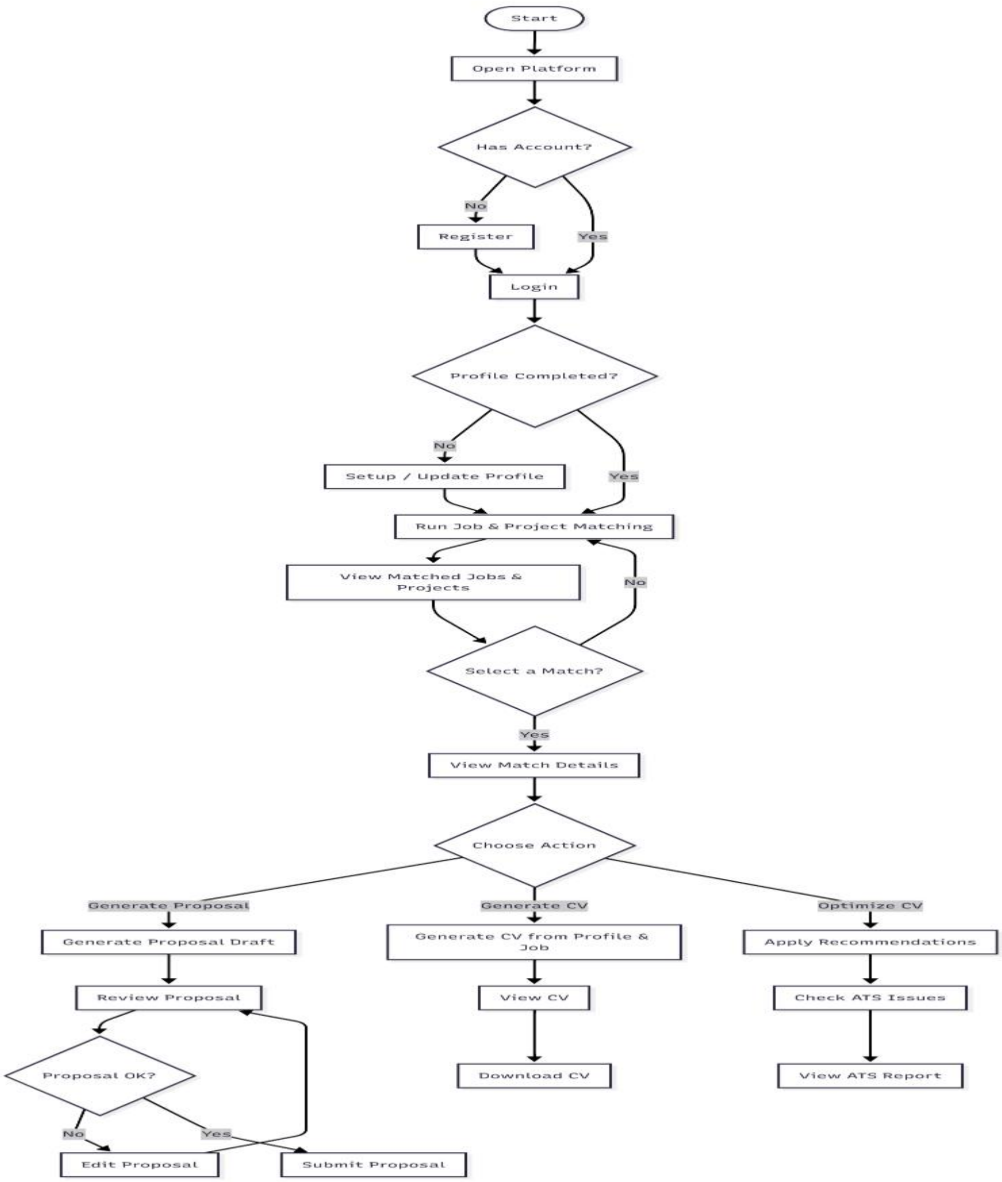
3.6 Use Case Diagram:



This Use Case Diagram represents the interactions between the primary system actors—USER (Job Seeker/Freelancer), Hiring/Freelancing Platforms, and ATS Providers—and the core functionalities of the SuperCareer AI platform. The use cases are structured into integrated functional modules, including Profile Management, Opportunity Discovery, AI Document Generation, and ATS Optimization.

The diagram highlights the system's ability to bridge the gap between candidate profiles and external market data through advanced back-end processes such as Job Matching Algorithms and Fine-Tuned LLMs. Each actor is associated with specific use cases that define their role in the ecosystem; while the User focuses on discovering opportunities and generating tailored materials, the external actors serve as data and validation providers. This outlines a seamless workflow where the system automates the scraping of projects and jobs, identifies ATS errors, and provides intelligent recommendations to enhance the user's competitive edge in the labor market.

3.7 User Activity Diagram:



This diagram illustrates the end-to-end journey of a job seeker or freelancer on the platform. It starts with the authentication phase, followed by a decision point regarding profile completeness. If the profile is ready, the system automatically triggers the matching engine. The user can then browse matched opportunities and choose between three primary AI-driven actions: generating a customized proposal, creating a tailored CV, or performing an ATS optimization check. This workflow ensures that the user is guided toward the most effective application strategy.

3.8 Sequence Diagram:

3.8.1 User Authentication and Profile Setup:

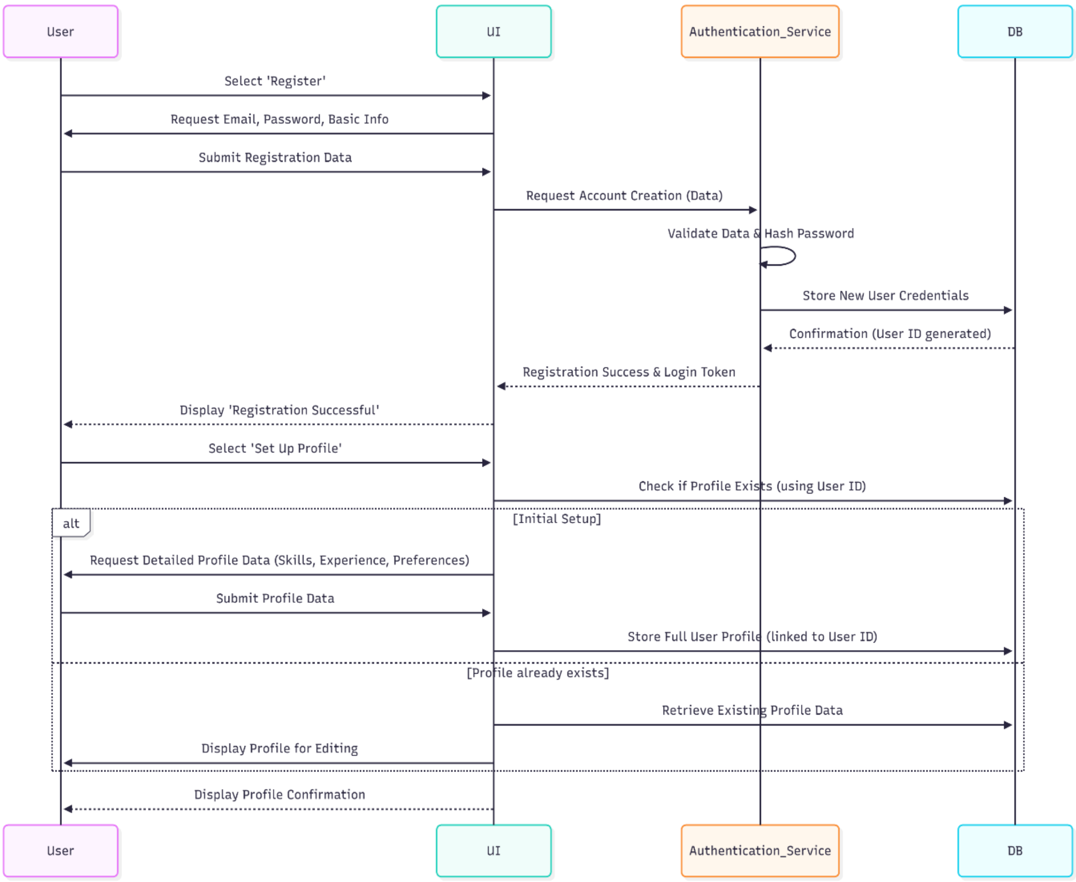
This sequence details the user onboarding process, from initial registration to creating the detailed profile needed for the AI matching services.

Components

- **User:** The human actor.
- **UI:** User Interface.
- **Auth Service:** Authentication and validation components.
- **DB:** Database.

Sequence Flow

1. The **User** submits registration data (email, password) via the **UI**.
2. The **UI** sends the data to Auth **Service** for account creation.
3. The **Auth Service** validates the data and **hashes the password** before storing the new credentials in DB.
4. Upon successful registration, the **User** is prompted to **Set Up Profile**.
5. The **UI** checks the **DB** to see if a profile already exists for the User ID.
6. *If Initial Setup:* The **UI** requests **Detailed Profile Data** (skills, experience, preferences) from the **User**.
7. The **UI** stores the complete **Full User Profile** in the **DB**.



3.8.2 Automated Data Scraping and Ingestion:

This sequence runs automatically to ensure the system has a constant supply of fresh job and project data.

Components

- **Scheduler:** The automated component that triggers the job.
- **Scraping Service:** Manages the external data requests.
- **Freelancing/Hiring Websites:** External data sources.
- **DB:** Database.

Sequence Flow

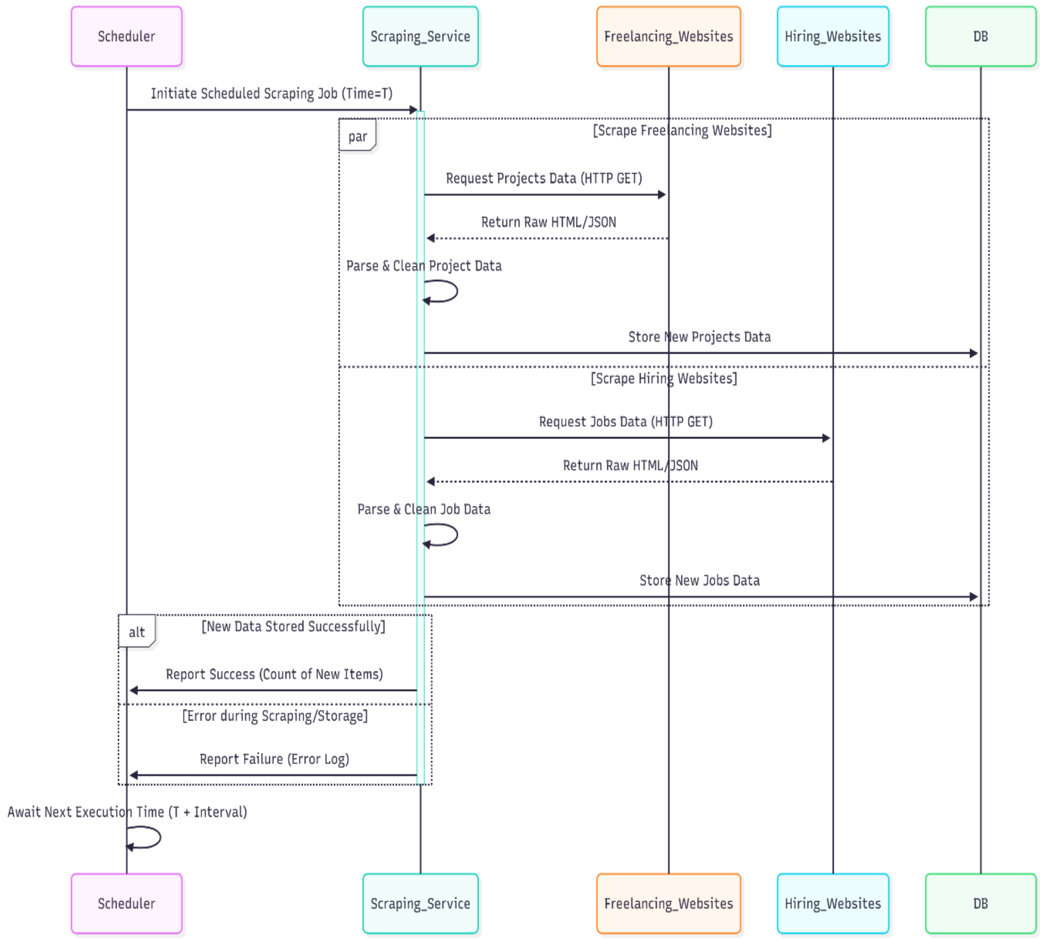
1. The **Scheduler** initiates the **Scraping Job** at a scheduled time (e.g., daily).

2. The **Scraping Service** sends parallel requests to **Freelance Websites** and **Hiring Websites** to retrieve raw data (HTML/JSON).

3. The **Scraping Service parses and cleans** the received raw data.

4. The cleaned data (**New Jobs/Projects Data**) is stored in DB.

5. The **Scraping Service** reports the outcome (success or error log) back to the **Scheduler**.



3.8.3 Job Matching and Notification:

This sequence describes how the AI core identifies relevant opportunities for the user and alerts them.

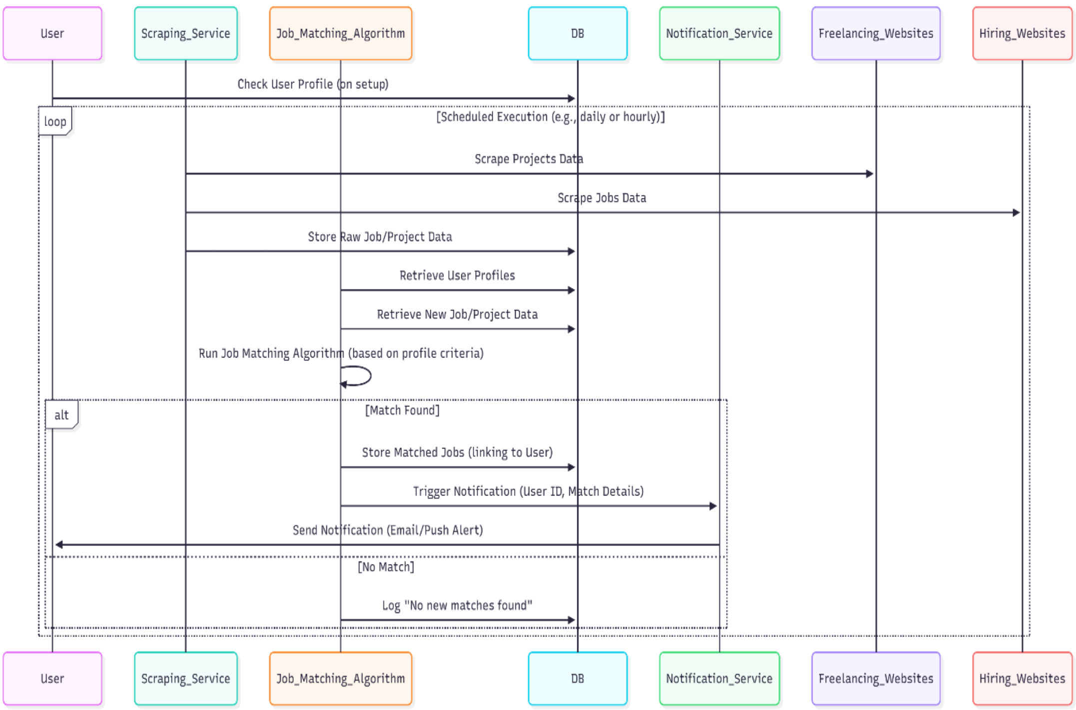
Components

- **Job Matching Alg.:** The core AI algorithm.

- **Notification Service:** Sends alerts (email/push).
- **DB:** Database.
- **User:** The recipient of the notification.

Sequence Flow

1. The **Job Matching Algorithm** is triggered (typically by the storage of new job data in the DB).
2. It retrieves **New Job Data** and **User Profiles** from DB.
3. The **Algorithm** runs its **Matching Logic** (comparing job requirements to user profile).
4. *If a Match is Found:*
   - The match is logged to the **DB**.
   - The **Notification Service** is triggered with the User ID and match details.
   - The **User receives the notification** (e.g., email or push alert).



3.8.4 Custom Proposal Generation:

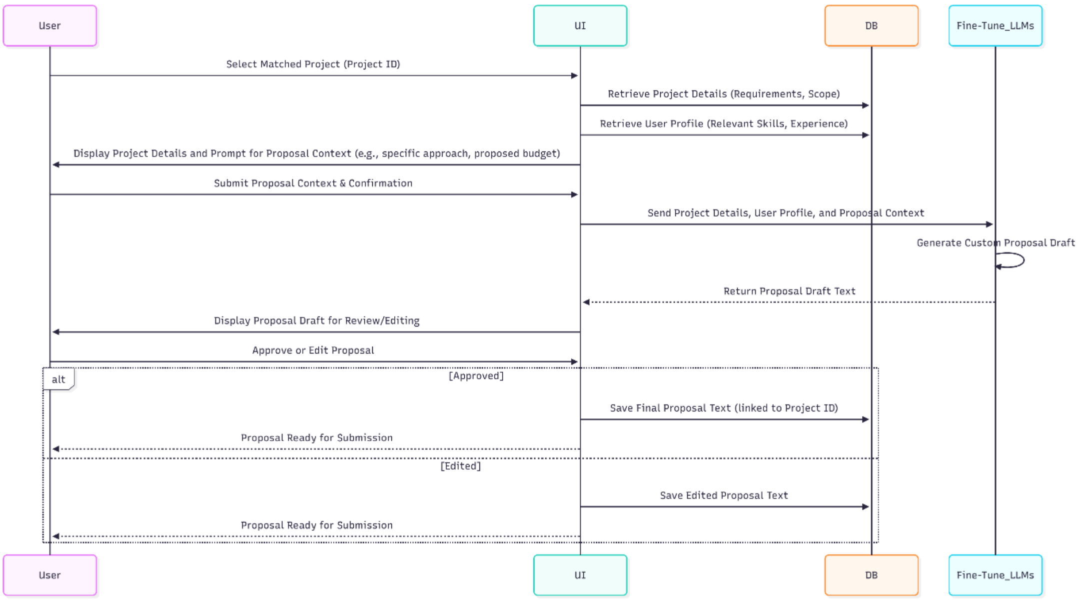
This sequence outlines the process where the user leverages the **Fine-Tune LLMs** to create a personalized proposal for a matched project.

Components

- **User:** The actor requesting the proposal.
- **UI:** User Interface.
- **DB:** Database.
- **Fine-Tune LLMs:** Large Language Models used for content generation.

Sequence Flow

1. The **User** selects a **Matched Project** on the **UI**.
2. The **UI** retrieves the **Project Details** and the **User Profile** from the **DB**.
3. The **User** provides **Proposal Context** (e.g., specific approach, budget) via the **UI**.
4. The **UI** sends all contextual data to the **Fine-Tune LLMs**.
5. The **LLMs** process the data and return a **Proposal Draft Text** to the **UI** (**Create Custom Proposal**).
6. The **User** reviews the draft and approves or edits it.
7. The final proposal text is **saved in the DB**.



3.8.5 CV Generation and Optimization:

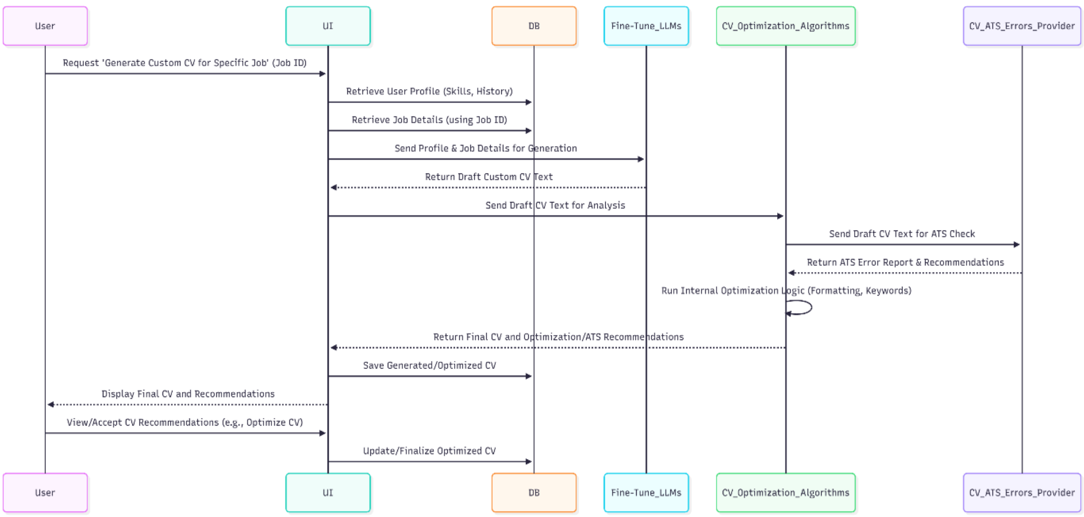
This sequence is the most complex, involving multiple AI/external services to generate, check, and optimize a custom CV for a specific job.

Components

- **Fine-Tune LLMs:** Generates the draft CV.
- **CV Optim. Alg.:** Analyzes and optimizes the CV based on internal rules.
- **ATS Provider:** External service for checking Applicant Tracking System errors.
- **DB:** Database.

Sequence Flow

1. The **User** requests **Generate Custom CV** for a specific Job ID via the **UI**.
2. The **UI** sends the User Profile and Job Details to **Fine-Tune LLMs**.
3. The **LLMs** return the **Draft CV Text** to the **UI** (**Generate Custom CV**).
4. The **UI** sends the draft to the **CV Optimization Algorithm** for analysis.
5. The **CV Optim. Alg.** sends the text to the external **ATS Provider** for a check (**Check ATS Issues**).
6. The **ATS Provider** returns an **ATS Error Report**.
7. The **CV Optim. Alg.** runs its **Internal Optimization Logic** (formatting, keywords).
8. The **CV Optim. Alg.** returns the **Optimized CV and Recommendations** to the **UI** (**Optimize CV**).
9. The **User** views and can accept the recommendations. The final, optimized CV is **saved in the DB**.



3.8.6 Admin Login and Dashboard Check:

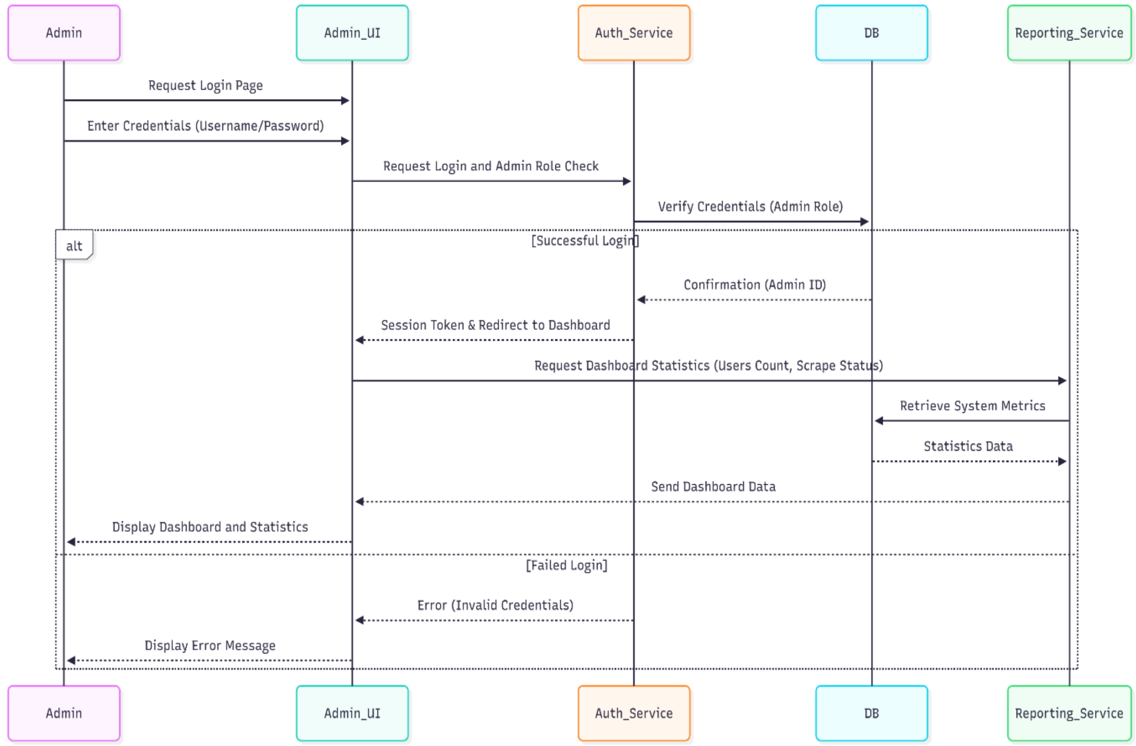
This flow validates the administrator's identity, grants elevated privileges, and loads the system dashboard with critical operational metrics.

1.Components

Component	Role in the Flow
Admin	The administrative user interacting with the system.
Admin UI	The secure, separate web interface for administrators.
Auth Service	Verifies credentials and checks for the mandatory Admin Role.
DB (Database)	Stores credentials, user roles, and system configuration.
Reporting Service	Aggregates and calculates real-time system metrics (e.g., active users, scraping success rates).

2.Sequence Flow Description:

1. **Request & Input :The Admin requests the login page and submits their credentials (Username/Password) to the Admin UI.**
2. **Authentication :The Admin UI forwards the credentials to the Auth Service to initiate a synchronous login request.**
3. **Authorization :The Auth Service queries the DB to verify the credentials and confirms that the user has the necessary Admin Role.**
4. **Success Path (Alt) :If successful, the Auth Service returns a Session Token to the Admin UI and initiates a redirect to the dashboard.**
5. **Data Loading :The Admin UI immediately requests essential Dashboard Statistics from the Reporting Service.**
6. **Reporting :The Reporting Service retrieves the necessary metrics from the DB ,compiles them, and sends the data back to the Admin UI for display.**
7. **Failure Path (Else) :If authentication fails, the Auth Service returns an error message, which the Admin UI displays to the Admin.**



3.8.7 Job Data Management:

This process enables the Admin to perform manual oversight on the job data ingested by the Scraping Service, ensuring quality and accuracy

.

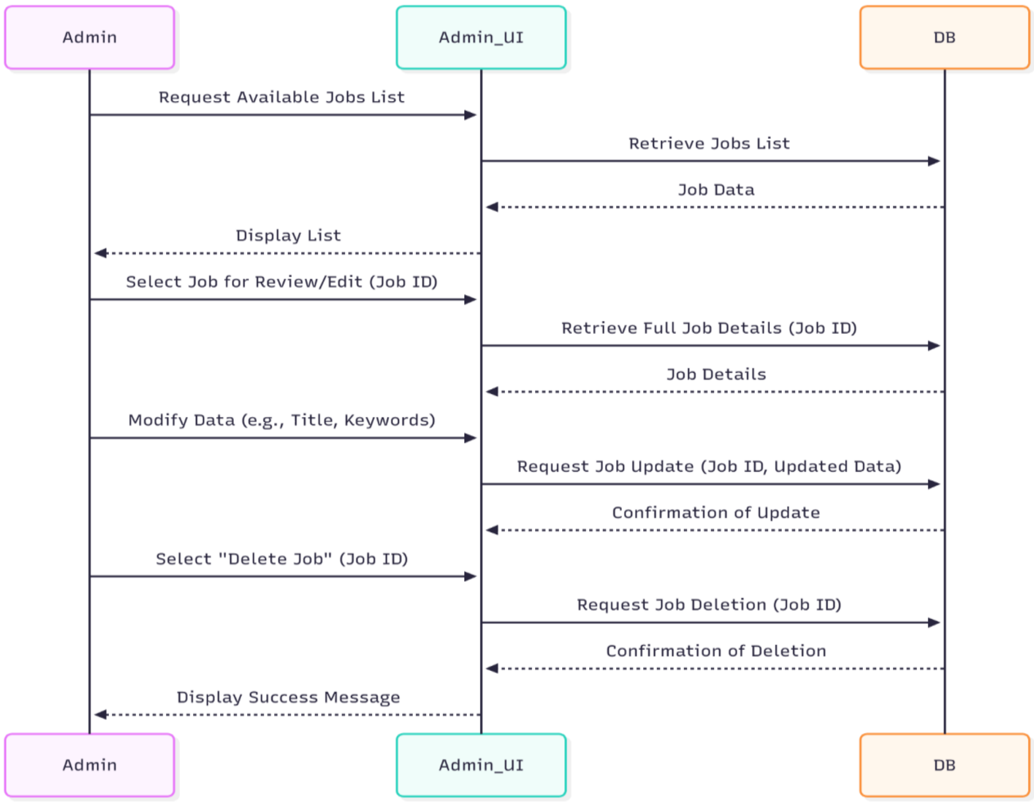
1.Components

Component	Role in the Flow
Admin	The administrator performing data quality checks.
Admin UI	Displays detailed job records and handles modification input.
DB (Database)	The repository for all scraped job data.

2.Sequence Flow Description

1. **Request List :The Admin requests to view the Available Jobs List via the Admin UI.**
2. **Retrieve Data :The Admin UI queries the DB to retrieve the list of jobs and displays it to the Admin.**
3. **Action Selection :The Admin selects a specific Job ID for review or modification.**
4. **Detail Retrieval :The Admin UI fetches the Full Job Details from the DB and displays all fields.**

5. **Modification:**
   - **The Admin edits the required fields (e.g., tags, keywords, source link).**
   - **The Admin UI sends a Job Update Request containing the updated data to the DB.**
   - **The DB confirms the successful update.**
6. **Deletion:**
   - **Alternatively, the Admin selects" Delete Job."**
   - **The Admin UI sends a Job Deletion Request to the DB.**
   - **The DB confirms the record deletion.**



3.8.8 User Management:

This flow is used for monitoring user activity, handling abuse, and managing user access rights within the system.

1,Components

Component	**Role in the Flow**
Admin	**The administrator responsible for moderation.**
Admin UI	**The interface used for displaying user records and executing status changes.**
DB (Database)	**Stores user profiles, status, and activity logs.**
Notification Service	**Used to communicate status changes (e.g., account block) to the user.**

2. Sequence Flow Description

1. **List Request :The Admin requests the Users List from the Admin UI.**

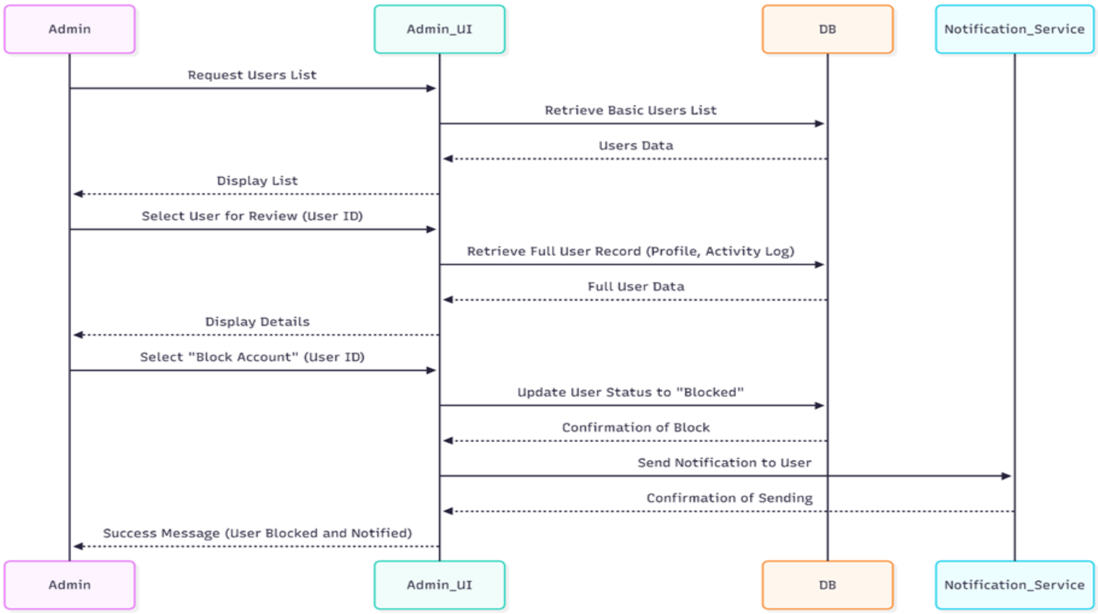
2. **Display List :The Admin UI retrieves the basic user information from the DB and displays the list.**

3. **Detailed Review :The Admin selects a specific User ID for review. The Admin UI fetches the Full User Record (including profile and activity log) from the DB.**

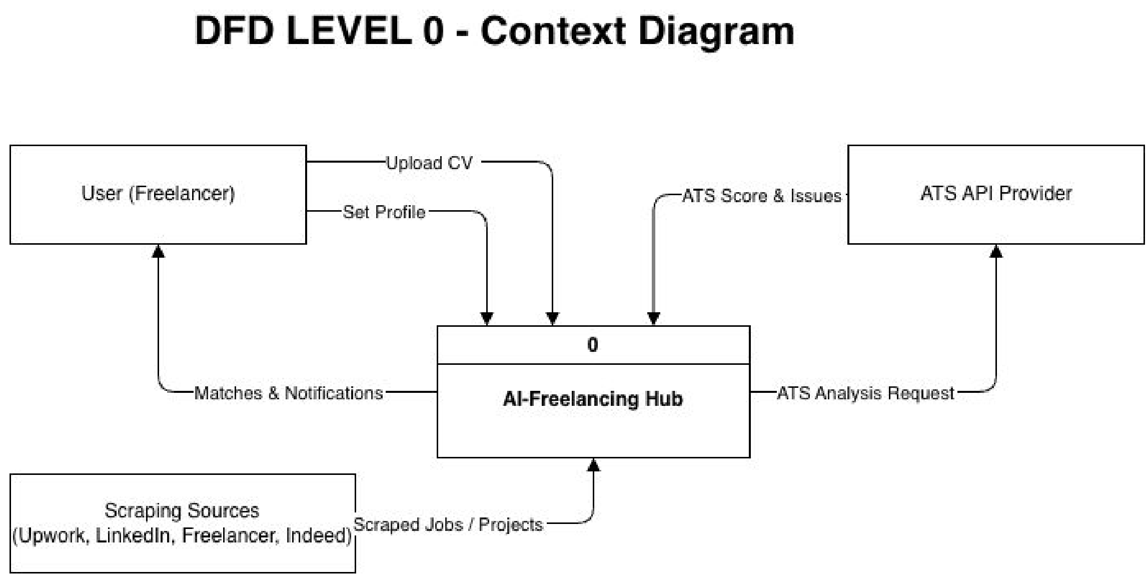
4. **Block/Unblock Action:**
   - **The Admin selects" Block Account) "or Unblock.(**
   - **The Admin UI sends a request to the DB to Update User Status to "Blocked".**
   - **The DB confirms the status update.**

5. **Notification :To maintain transparency, the Admin UI triggers the Notification Service to Send Notification to the affected user about the status change.**

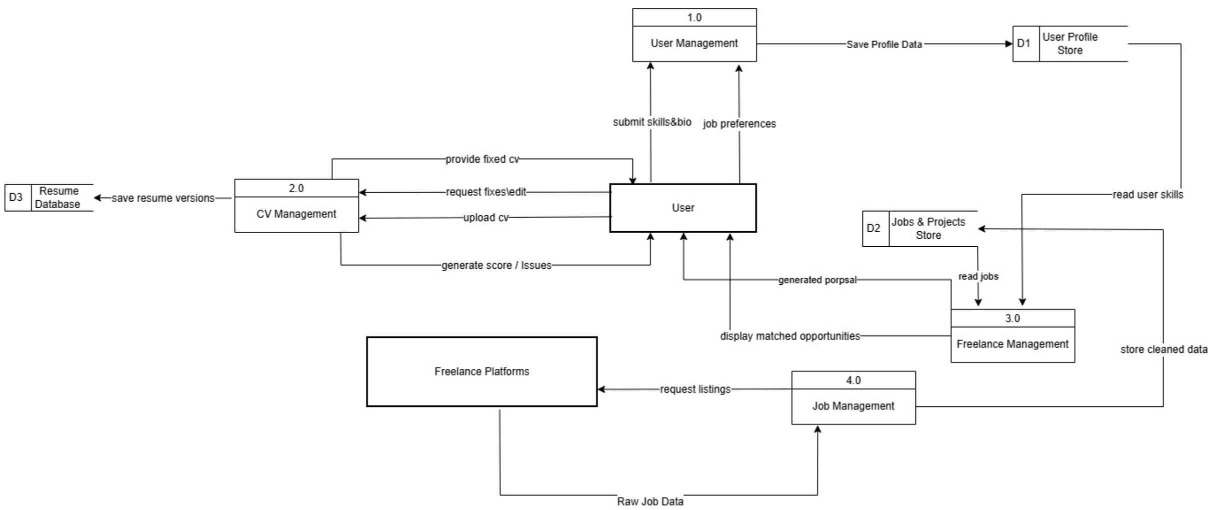
6. **Confirmation :The Notification Service confirms the send status, and the Admin UI presents a final success message to the Admin.**



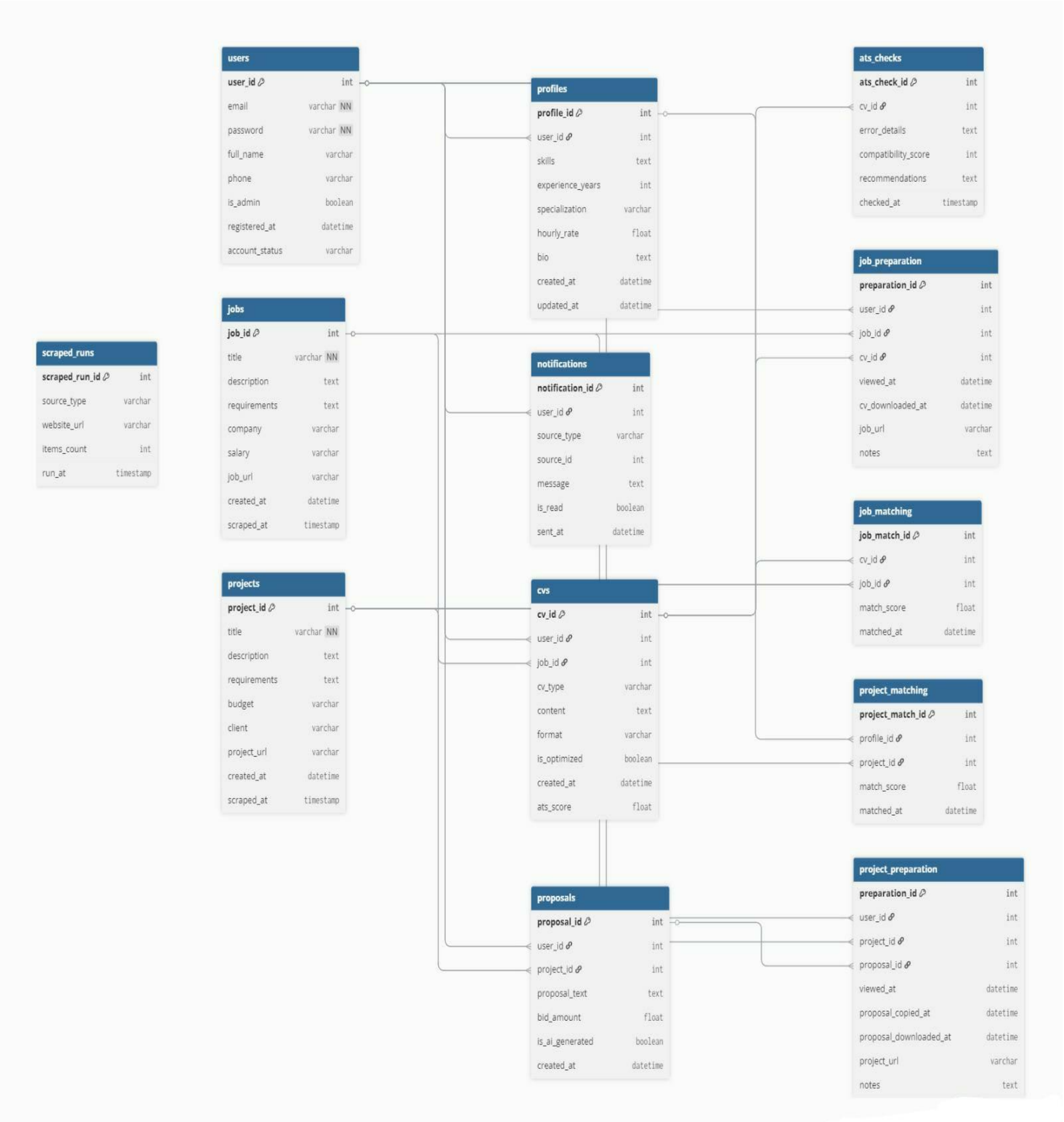
3.9 Context diagram:



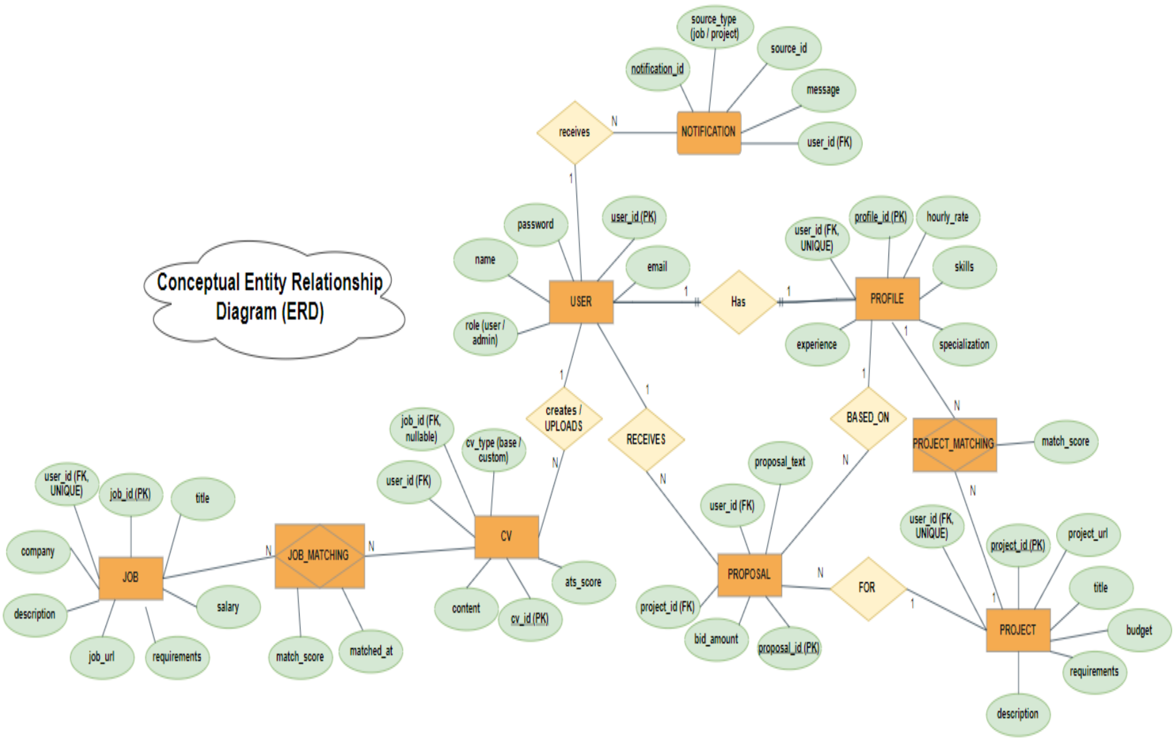
3.9.1 Data flow diagram:



3.9.3 Data Base Schema:



3.9.2 ERD:


Conceptual Entity Relationship Diagram (ERD)
NOTIFICATION
notification_id
source_type (job / project)
source_id
message
user_id (FK)
receives
USER
password
user_id (PK)
name
email
role (user / admin)
Has
PROFILE
user_id (FK, UNIQUE)
profile_id (PK)
hourly_rate
skills
experience
specialization
creates / UPLOADS
RECEIVES
BASED_ON
PROJECT_MATCHING
match_score
CV
job_id (FK, nullable)
cv_type (base / custom)
user_id (FK)
content
cv_id (PK)
ats_score
JOB_MATCHING
match_score
matched_at
JOB
user_id (FK, UNIQUE)
job_id (PK)
title
company
description
salary
job_url
requirements
PROPOSAL
proposal_text
user_id (FK)
project_id (FK)
bid_amount
proposal_id (PK)
FOR
PROJECT
user_id (FK, UNIQUE)
project_id (PK)
project_url
title
budget
requirements
description